An offline AI co-pilot that predicts MPLS network failures before they happen and explains every recommendation in plain English — grounded in your own runbooks, with zero internet access, ever.

Existing Solution Landscape (The Actual gap):

Cloud AIOps platforms (Splunk ITSI, Moogsoft, Datadog AI) — Powerful, but the AI/ML inference layer runs on vendor cloud servers.

NMS tools (Nagios, Zabbix, SolarWinds) :No prediction, no natural-language reasoning of why system failed

Enterprise on-prem AIOps suites : unexplainable AI recommendation can't be trusted or audited

WHAT WE OFFER

1.Fully air-gapped architecture

2.Predictive failure analysis

3.RAG-based, verifiable answers with direct runbook citations.

4.Comprehensive, transparent audit trails for all AI decisions.

5.Vendor-neutral, open-component stack (Ollama, ChromaDB, FastAPI) for full auditability.

6.Intuitive, natural-language conversational interface.

7.Action-oriented, prescriptive guidance rather than raw data logs.

USP:

Zero cloud dependency | **RAG-retrieved runbooks.** | **Predictive Intelligence** | **Fully Auditable** | **Vendor-Neutral**

FEATURES

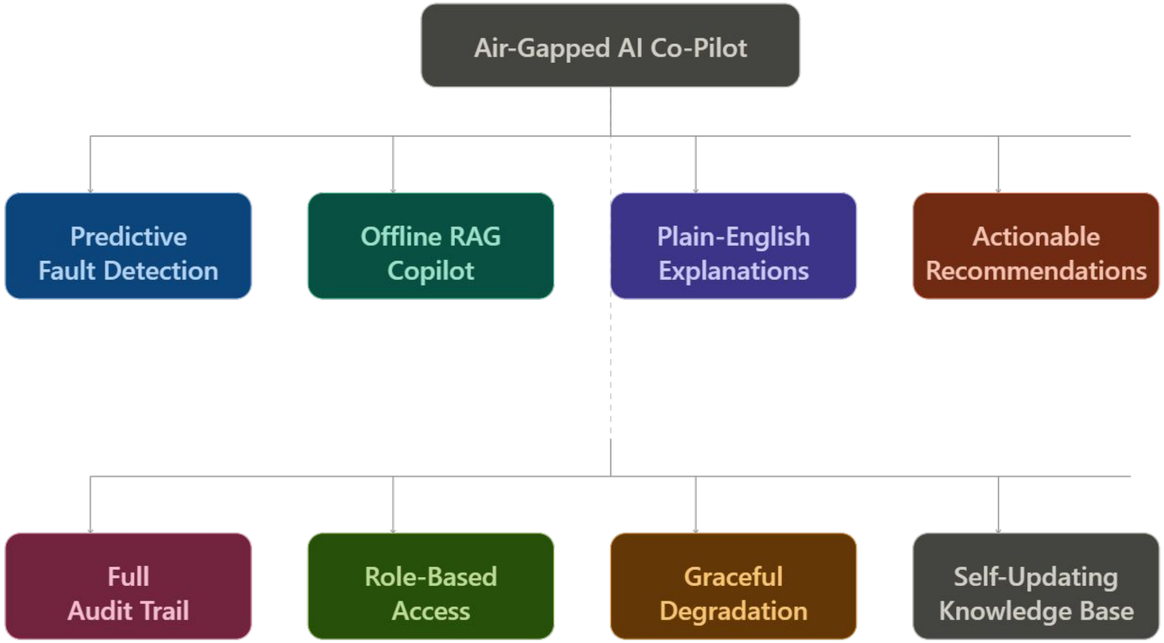
Predictive Fault Detection — flags anomalies in SNMP/syslog patterns and predicts failures before they cause downtime

Offline RAG-Powered Copilot — conversational Q&A interface grounded in your own runbooks and playbooks, zero internet required

Plain-English Explanations — converts raw log noise into clear root-cause reasoning, no protocol-level jargon

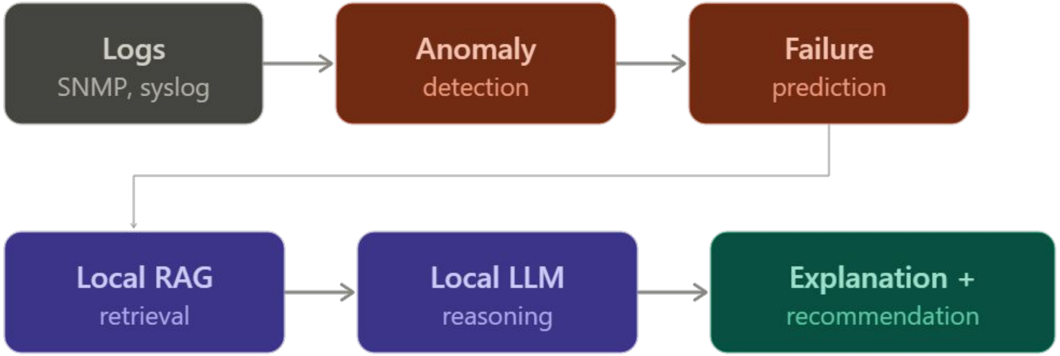
Actionable Recommendations — every suggestion cites the exact runbook section it came from, with a confidence score

Full Audit Trail — every prediction, retrieval, and recommendation is logged and timestamped for compliance review

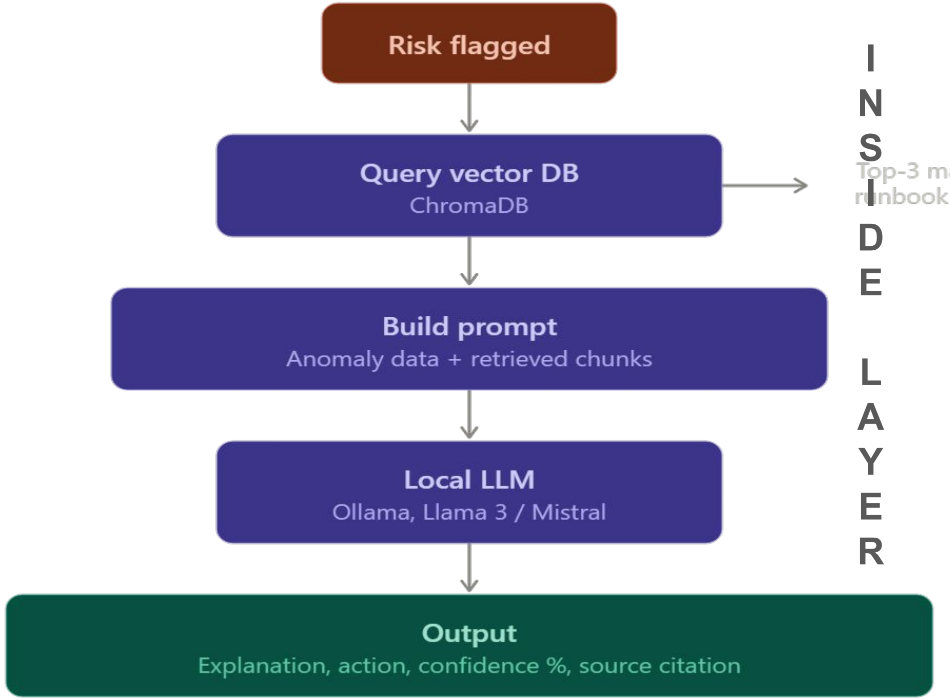


AI FEATURES RUN FULLY OFFLINE-NO INTERNET REQUIRED

Process Flow



End-to-End Process Flow



Key Inputs & Uniqueness

Zero internet
Every step runs fully on-prem

Cited sources
Every output cites its runbook

Predictive
Flags risk before failure, not after

Grounded only
No black-box guessing, ever

WIREFRAME

Air-Gapped AI Co-Pilot
NOC MONITORING SYSTEM V2.4.1

Offline mode active — no network egress
2026-06-29 10:43:02 UTC

NETWORK HEALTH	ACTIVE ALERTS	PREDICTIONS TODAY	AVG CONFIDENCE
87%	3	14	79%

PREDICTED FAILURES

Router-7	ETA ~18 min	82%
Switch-3	ETA ~47 min	61%
Link-A2	ETA ~2 hr	34%

ANOMALY TREND — ROUTER-7 CPU

LIVE

LIVE LOG STREAM

10:42:03	WARN	SNMP trap: link down on Gi0/1
10:42:11	INFO	BGP peer 192.168.1.1 state → Active
10:42:28	CRIT	Router-7 CPU threshold breached: 94%
10:42:41	INFO	Interface Gi0/2 duplex mismatch detected
10:42:55	WARN	Switch-3 memory utilization at 88%

COPILOT — INCIDENT #482

LOCAL MODEL

Why is Router-7 at risk?

CPU sustained above 90% for 12 minutes, matching a known BGP flap pattern. Historical data shows 3 prior incidents with identical CPU signature preceding interface failure.

Runbook #14 | Incident #391

What"s the recommended action?

Based on the BGP flap pattern, a session restart should stabilize CPU utilization without causing a full outage. Maintenance window not required for this action.

• SUGGESTED ACTION

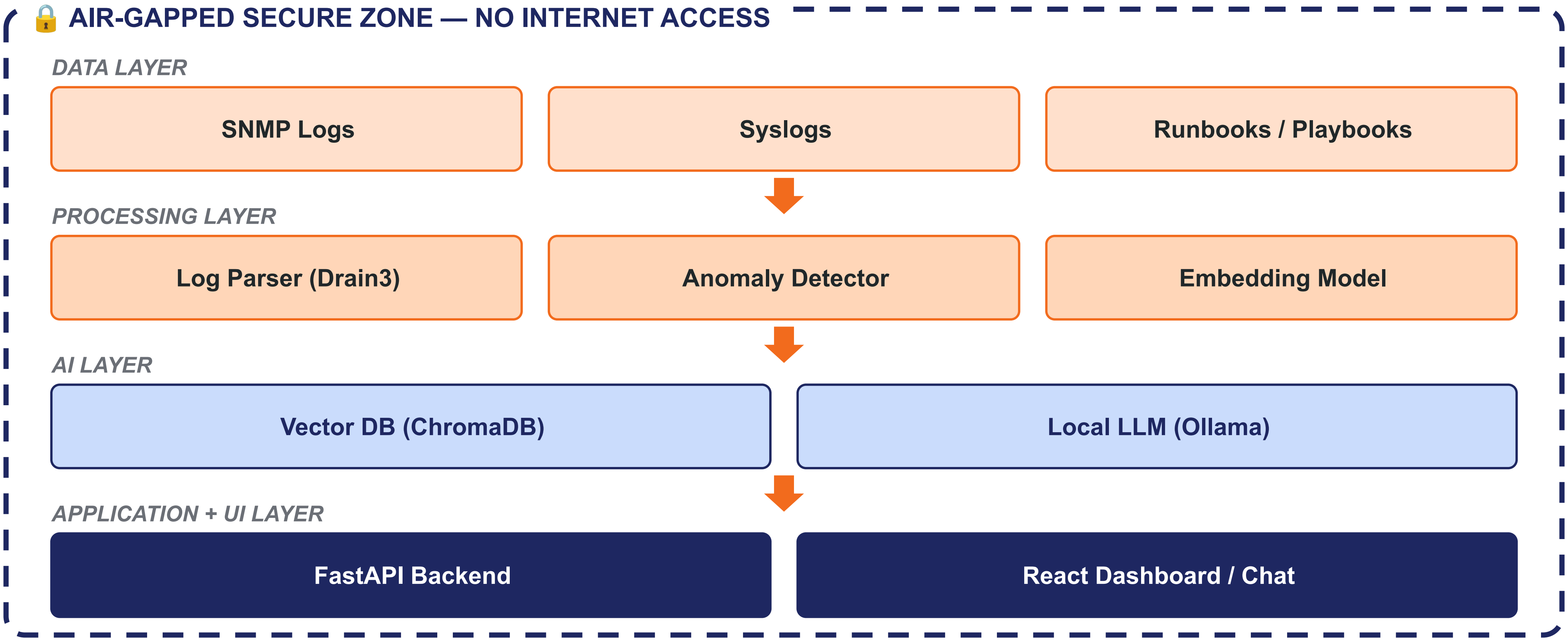
Restart BGP session on Router-7

Confidence: 82% · Est. downtime: 0s

Execute | Dismiss

Ask a follow-up...

System Architecture



Frontend:

React: Component-based architecture for reusable, testable UI; enables rapid development with zero ramp-up.

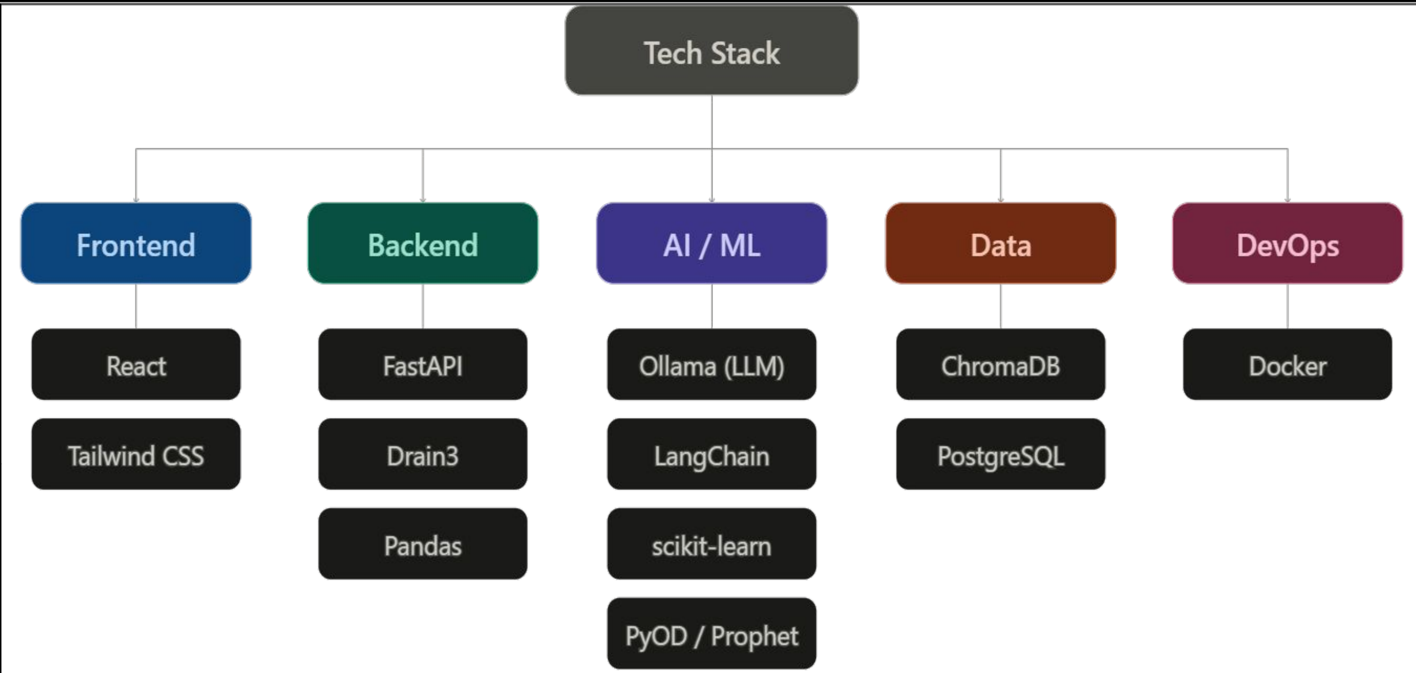
Tailwind CSS: Utility-first styling for fast, consistent dashboard and chat interfaces without custom CSS.

Backend Infrastructure:

FastAPI: Async framework for responsive LLM inference, supporting native Python ML/RAG pipelines.

Drain3: Real-time algorithm for parsing unstructured logs into consistent templates.

Pandas: Standard library for log data manipulation and time-series feature engineering.



AI / ML Stack

Ollama: Runs open models (Llama 3, Mistral) locally for air-gapped compliance; provides a simple API.

LangChain: Orchestrates the RAG pipeline, accelerating development and reducing integration risk.

scikit-learn: Provides battle-tested anomaly detection algorithms like Isolation Forest for quick deployment.

PyOD / Prophet: PyOD handles outlier detection; Prophet enables time-series forecasting to predict future failure trends.

Data

ChromaDB: Python-native vector database for offline runbook embeddings; enables high-accuracy similarity search for RAG.

PostgreSQL: ACID-compliant relational storage for incidents and logs; ensures reliable data relationships and audit compliance.

DevOps

Using Docker to containerize the entire stack.

Enables reproducible, air-gapped deployment on a single on-prem ISRO server, ensuring full system portability without cloud or internet reliance.

Estimated Implementation Cost

Category	Item	Estimated Cost
Compute Hardware	On-prem server (8-16GB RAM, optional GPU) - one-time	Rs 80,000 - 2,00,000
AI Models	Local LLM (Llama 3 / Mistral) + embeddings - open-source	Free (open-source license)
Software Stack	FastAPI, React, ChromaDB, PostgreSQL, Docker	Free (open-source)
Engineering Effort	4-person team, ~35-40 dev-hours for MVP	Team time (hackathon)
Maintenance	Offline update cycles via signed packages	Low - no per-API cloud billing